

2. **Установленный на момент останова режим функционирования КС / Mode of the CS operation during shutdown:**
В работе ГПА / Operated GCU № 2,3;
В резерве ГПА / Standby GCU №.
3. **Параметры ГПА согласно / Operating data of GCU according: Донесение № 2 / Report № 2 от / from**
14.04.2023.
4. **Дата и время останова ГПА / Date and time of the GCU shutdown:** 14.04.2023. 16:52.
5. **Состояние ГПА в момент останова / GCU condition during shutdown:**
В режиме работы в трассу / operating online.
(в режиме работы в трассу, холодная прокрутка, подготовка к запуску / operating online, dry crank, preparation to start)
6. **Обстоятельства, при которых произошел останов (принятые меры для восстановления работы режима в трассу) / Circumstances during which the shutdown has occurred (actions taken to recover operating mode online):** Во время работы ГПА №2 произошел сбой передачи данных от датчиков системы минерального масла LIT-174, PIT-111, PDIT-109, PIT-186, TE-180 (приложение №1, фото-2) а также датчиков системы сухого газового уплотнения PIT-750, PDIT-765, PIT-765B, PIT-757B, что повлекло за собой Автоостанов ГПА №2. Причиной сбоя передачи данных послужил выход из строя модуля изолятора Profibus (Railbus MTL) тэговый номер RB6-2 (Приложение №1 фото-4) установленной в шкафу RIO Panel JB-2./
During the operation of GCU No. 2, there was a failure in the transmission of data from the sensors of the mineral oil system LIT-174, PIT-111, PDIT-109, PIT-186, TE-180 (Appendix #1, photo-2) as well as the sensors of the dry gas seal system PIT-750, PDIT-765, PIT-765B, PIT-757B, which resulted in automatic shutdown of GCU No. 2. The reason for the failure of data transmission was the failure of the Profibus (Railbus MTL) isolator module tag #RB6-2 (Appendix #1 photo-4) installed in the RIO Panel JB-2 cabinet./
7. **Поврежденные узлы, детали, приборы / Damaged units, details, devices:**
Модуль изолятор Railbus (MTL 8000 I/O 8922-RB-IS) установленной в шкафу RIO Panel JB-2./
/ isolator module (MTL 8000 I/O 8922-RB-IS) tag #RB6-2 installed in the RIO Panel JB-2 cabinet
8. **Принятые меры для устранения причин останова / Measures performed to eliminate the reasons of shutdown:**
После выявления и замены неисправного модуля изолятора Railbus тэговый номер RB6-2 в шкафу RIO Panel JB-2, запустили ГПА №2 заново / After identifying and replacing the faulty Railbus isolator module tag #RB6-2 in the RIO Panel JB-2 cabinet, GCU No. 2 was restarted /
9. **Заключение (пригодность восстановленного ГПА к дальнейшей эксплуатации) / Conclusion (applicability of the recovered GCU for further operation):**
ГПА №2 выведен в «резерв»/ GCU #2 was put into the "reserve" mode.

10. Мероприятия по недопущению аналогичного автоматического останова / Actions taken to prevent similar auto shutdown:

Заменить оборудование установленные более 5 лет назад так как электронные компоненты выработают за длительный промежуток времени свой ресурс. Вышедши из строя изолятор Railbus 2009 года выпуска и проработал более 10 лет./ Replace equipment installed more than 5 years ago, as electronic components will develop their resource over a long period of time. The railbus insulator was built in 2009 and has been in operation for over 10 years.

11. Примечание / Notes: Системах минерального масла и сухого газового уплотнение, датчики влияющие на режим работы ГПА имеют резервирование за исключением датчика давления третичного уплотнение (тэговый номер PIT-750). При выходе из строя модуля на катаром подключен (PIT-750) или выходе из строя изолятора Railbus происходит авто останов ГПА. В целях обеспечения надежности передачи данных рекомендуется установить резервный датчик давления с обновлением алгоритма САУ ГПА./ In mineral oil and dry gas seal systems, the sensors that affect the operation mode of the GCU are redundant, with the exception of the tertiary seal pressure sensor (tag number PIT-750). If the module on the Qatar is connected (PIT-750) or the Railbus isolator fails, the GCU will automatically stop. In order to ensure the reliability of data transmission, it is recommended to install a backup pressure sensor with an update of the GCU ACS algorithm.

12. Приложение / Attachment: Приложение №1/ Attachment #1

(фото, графики и т.д. / photos, diagrams, etc.)

Фото-2 сбой передачи данных системы минерального масла / communication failure of the mineral oil system

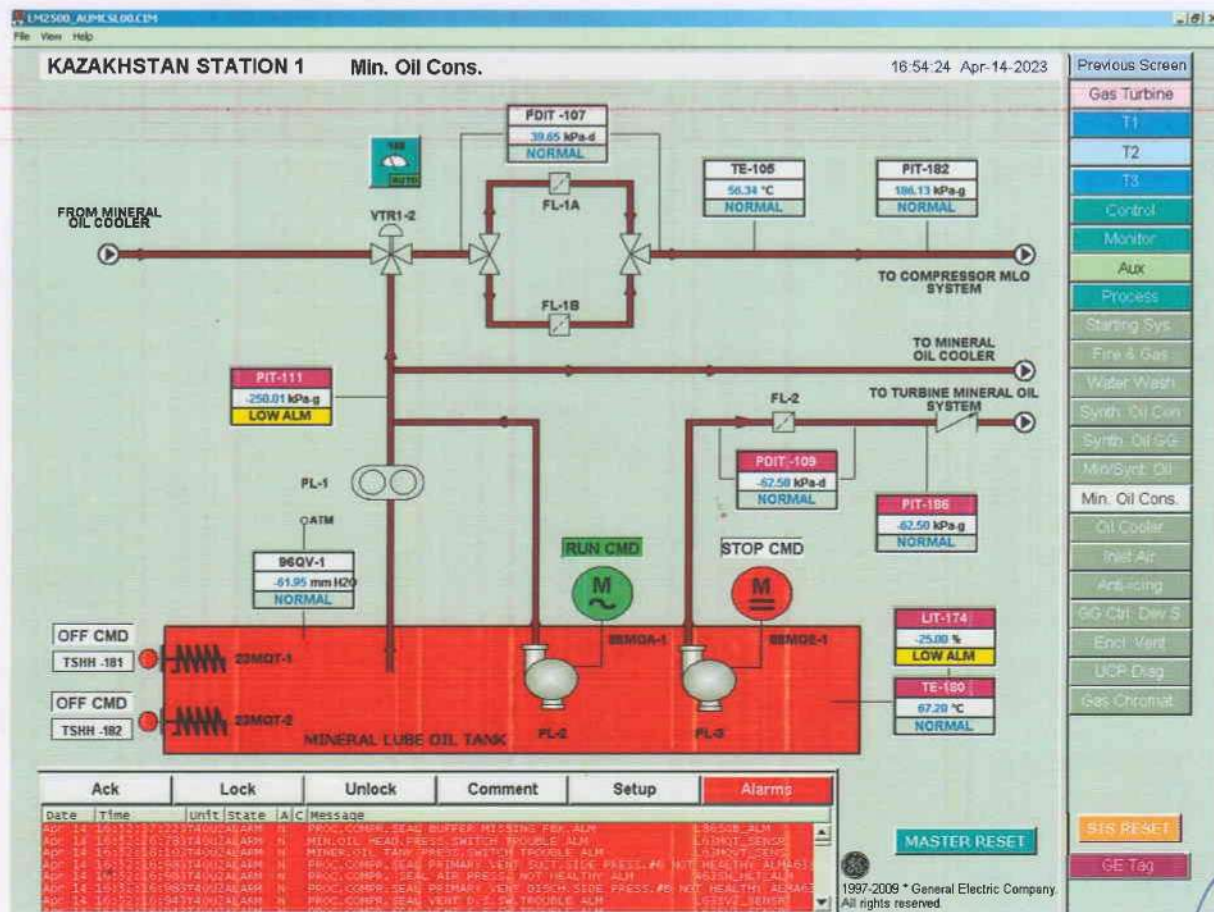


Фото -3 сбой передачи данных системы уплотнение / communication failure of the seal system

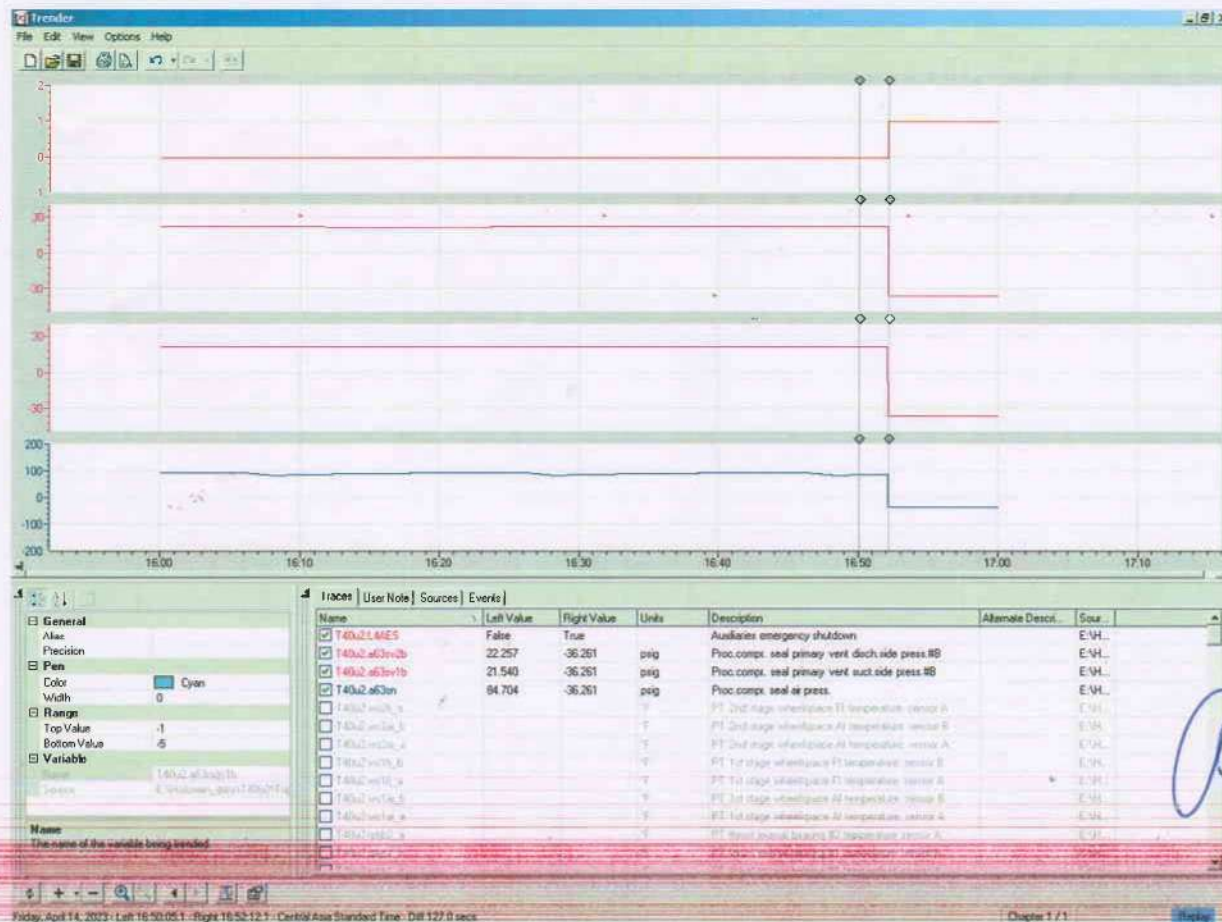


Фото-1 Журнал ошибок/ Alarms list

Alarm Screen

KAZAKHSTAN STATION 1

ALARMS

18:56:00 Apr-14-2023

358 Apr 14 16:55

Date	Time	Unit	Alarm ID	State	A	C	Message
04/14/23	16:55:30:183	T4002	T4002_P1679...	NORMAL	N		UCP 2 Bently Nevada Vibration Monitor Fault 186vmbn
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.SEAL BUFFER MISSING FEEDB. ALM L689N ALM
04/14/23	16:55:30:183	T4002	T4002_P1680...	ALARM	N		MIN.OIL HEAD PRESS.SWITCH TROUBLE ALM L689N SENSE
04/14/23	16:55:30:183	T4002	T4002_P1680...	ALARM	N		MIN.OIL TANK PRESS.SWITCH TROUBLE ALM L689N SENSE
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.SEAL PRIMARY VENT DISCH.SIDE PRESS.#8 NOT HEALTHY ALMA33VIB
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		PROC.COMPR. SEAL AID PRESS. NOT HEALTHY ALM L689N HLT ALM
04/14/23	16:55:30:183	T4002	T4002_P1680...	ALARM	N		PROC.COMPR.SEAL PRIMARY VENT SUCT.SIDE PRESS.#8 NOT HEALTHY ALMA33VIB
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.SEAL VENT DISCH.SIDE TROUBLE ALM L689N SENSE
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.SEAL VENT DISCH.SIDE TROUBLE ALM L689N SENSE
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.THRUST BRG TEMP ACTIVE SIDE NOT HEALTHY ALMA33VIB HLT ALM
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		MINERAL OIL HEADER TEMPERATURE #8 NOT HEALTHY ALM L689N HLT ALM
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		MINERAL OIL TANK TEMPERATURE NOT HEALTHY ALM L689N HLT ALM
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.JOURNAL BEARING #8R TEMPERATURE NOT HEALTHY ALMA33VIB HLT ALM
04/14/23	16:55:30:183	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.JOURNAL BEARING #8R TEMPERATURE NOT HEALTHY ALMA33VIB HLT ALM
04/14/23	16:55:30:183	T4002	T4002_P1680...	ALARM	N		MIN.OIL PUMPS DISCH PRESS. LOW ALARM L689N ALM
04/14/23	16:55:30:183	T4002	T4002_P1680...	ALARM	N		MINER.OIL TANK LEVEL LOW ALARM L689N ALM
04/14/23	16:52:06:023	T4002	T4002_P1680...	NORMAL	N		PROC.COMPR.ROT BYPASS MISSING FEEDB. ALM L689N ALM
04/14/23	16:52:07:943	T4002	T4002_P1681...	NORMAL	N		Proc. compr. antisurge valve mismatch feedback - ALMA20AS FLT
04/14/23	16:52:07:943	T4002	T4002_P1681...	ALARM	N		Four hours not-actuated sequence completed active - ALMA20AS FLT
04/14/23	16:52:07:943	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.SEAL AIR PRESS. LOW TRIP L689N ALM
04/14/23	16:52:07:943	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.SEAL AIR PRESS. LOW TRIP L689N ALM
04/14/23	16:52:07:943	T4002	T4002_P1679...	ALARM	N		MINERAL OIL TANK LEVEL NOT HEALTHY ALM L689N HLT ALM
04/14/23	16:52:07:943	T4002	T4002_P1679...	ALARM	N		MINERAL OIL DISCHARGE PUMPS PRESSURE NOT HEALTHY ALMA33VIB HLT ALM
04/14/23	16:52:07:943	T4002	T4002_P1679...	ALARM	N		MINERAL OIL EMERGENCY FILTER DIFF. PRESS. NOT HEALTHY ALMA33VIB HLT ALM
04/14/23	16:52:07:943	T4002	T4002_P1679...	ALARM	N		MINERAL OIL EMERGENCY PUMP HEADS PRESS. NOT HEALTHY ALMA33VIB HLT ALM
04/14/23	16:52:07:943	T4002	T4002_P1679...	ALARM	N		MINERAL OIL TANK PRESSURE #8 NOT HEALTHY ALM L689N HLT ALM
04/14/23	16:52:07:943	T4002	T4002_P1679...	ALARM	N		PROC.COMPR. SEAL INLET/BALANC. DIFF. PRESSURE NOT HEALTHY ALMA33VIB HLT ALM
04/14/23	16:52:07:943	T4002	T4002_P1679...	ALARM	N		MINERAL OIL HEADER PRESSURE #8 NOT HEALTHY ALM L689N HLT ALM
04/14/23	16:48:53:983	T4002	T4002_P1679...	NORMAL	N		PROC.COMPR.SEAL INLT./BAL. PRESS. LOW ALM L639GLA ALM
04/14/23	16:48:53:983	T4002	T4002_P1679...	ALARM	N		CRK T/M Hall Effect - 10 mA C 248DHALM
04/08/23	21:42:44:543	T4002	T4002_P1679...	ALARM	N		PROC.COMPR.SEAL VENT S.S.FLOW LOW ALM L689VIL ALM
04/08/23	21:42:44:543	T4002	T4002_P1680...	ALARM	N		48V FAILURE ALARM FLAG C 248DHALM
04/08/23	21:42:44:543	T4002	T4002_P1680...	ALARM	N		48V FAILURE ALARM FLAG C 248DHALM
04/08/23	21:42:44:543	T4002	T4002_P1680...	ALARM	N		48V FAILURE ALARM FLAG C 248DHALM
01/13/23	22:10:45:466	T1002	T1002_P1680...	NORMAL	N		PX36 High Acoustics Alarm Fals C PX36HALM
12/13/22	16:21:30:286	T1002	T1002_P1679...	NORMAL	N		T48C sensor failed alarm C T48CALM
12/13/22	16:08:01:606	T1002	T1002_P1679...	NORMAL	N		T48G sensor failed alarm C T48GALM
12/12/22	17:12:36:576	T1002	T1002_P1680...	NORMAL	N		Flame detector A failed alarm C FLAMEALM
12/12/22	09:25:09:886	T1002	T1002_P1680...	NORMAL	N		NGG difference alarm flag C NGGDHALM
11/26/22	20:01:40:626	T1002	T1002_P1679...	NORMAL	N		AnalogInputHighErr. PCN01_dev1_AnalogInpu
11/26/22	20:01:38:056	T1002	T1002_P1680...	NORMAL	N		CDP T/M open or short (R Coil) CDPINOS_R
11/26/22	20:01:38:056	T1002	T1002_P1680...	NORMAL	N		VSV T/M open or short (R Coil) VSVTR_R
11/26/22	20:01:37:626	T1002	T1002_P1679...	NORMAL	N		AnalogInputLowErr. PCN01_dev1_AnalogInpu
10/06/22	11:57:13:004	T4002	T4002_P1680...	ALARM	N		AIR FILTER PULSE VET SYSTEM FAULT ALARM L689N ALM
09/17/22	15:00:35:401	T1002	T1002_P1680...	NORMAL	N		T48 sensor G difference alarm flag C T48GHALM
09/02/22	19:11:17:454	T1002	T1002_P1679...	ALARM	N		StartupPositionSensorErr. PCN01_dev1_StartupPos
09/02/22	19:11:17:454	T1002	T1002_P1679...	ALARM	N		StartupPositionSensorErr. PCN01_dev1_StartupPos
09/02/22	19:11:17:454	T1002	T1002_P1679...	ALARM	N		StartupPositionSensorErr. PCN01_dev1_StartupPos

Ack

Ack All

Lock

Unlock

Comments

Exit

Ack

Ack All

Lock

Unlock

Comments

Exit

Фото-4 неисправного модуля изолятора Railbus в шкафу RIO Panel JB-2/ Photo of failed Railbus Isolator Module in RIO Panel JB-2 cabinet



[Handwritten signature]